

Jason Xu

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PROFESSIONAL SUMMARY

Bioinformatician and Sr. Programmer Analyst profile targeted to Columbia's Center for Infection and Immunity, with 4+ years of combined computational biology, machine-learning, and scientific software experience across biotechnology, academic research, AI evaluation, and federal analytics settings. M.S. Bioinformatics candidate with a B.S. in Computer Science and hands-on strengths in microbial detection and discovery, host-response RNA-seq, high-throughput serology and protein-array-style assay analysis, microbiome and multi-omics integration, reproducible pipelines, database-backed reporting, statistical/ML methods, publications, and research grant-ready scientific communication.

TECHNICAL SKILLS

Infection & Immunity Bioinformatics: microbial surveillance, pathogen detection/discovery, host-response profiling, high-throughput serology, protein arrays, antibody-antigen assays, microbiome analysis, viral particle assays, diagnostic assay QC, autism and chronic fatigue syndrome/myalgic encephalomyelitis study support

Multi-Omics & Genomics: RNA-seq, differential expression, functional genomics, proteomics/metabolomics integration concepts, pathway analysis, motif discovery, sequence alignment, variant/nucleic-acid detection, DESeq2, edgeR, limma, MEME Suite

Programming & Data Systems: Python, R, SQL, Bash, pandas, NumPy, Bioconductor, ggplot2, PostgreSQL/MySQL, metadata schemas, data cleaning, database-ready outputs, reproducible reports

Pipelines & Infrastructure: FastQC, STAR, featureCounts, BWA, BLAST, GATK, Snakemake, Nextflow, Docker, Conda, Linux, Git, HPC/SLURM, workflow documentation, CLIA-style validation support

Statistics & Machine Learning: statistical modeling, QC metrics, model evaluation, feature engineering, scikit-learn, PyTorch, transformers, RAG, anomaly/failure-mode detection, interpretation of biological signals

EDUCATION

Boston University | M.S. Bioinformatics; Genomics Data Science, Statistical Modeling, Biological Databases, Machine Learning for Biomedical Data | Boston, MA | Expected Dec. 2026

University of Washington | B.S. Computer Science, Dean's List | Seattle, WA | 2025

EXPERIENCE

Bioinformatics Analyst Intern, Full-Time | IRIS Kinetics | *Boston, MA* | 2026-Present

- Develop and maintain Python/R workflows for high-throughput, label-free multiplexed microarray biosensing data, including sample metadata validation, batch-level QC, normalization-ready outputs, visualization, and documented handoff for translational assay teams.
- Analyze protein-array-style and high-throughput serology-relevant datasets spanning antibody-antigen interactions, viral particle assays, nucleic-acid/SNP detection, microRNA/protein biomarkers, and viral vector QC, translating results into scientist-facing biological summaries.
- Support microbial detection and diagnostic-assay development use cases by reviewing sensitivity trends, capture-probe performance, kinetic binding curves, association/dissociation patterns, off-rate behavior, and assay reproducibility across experimental runs.
- Create database-ready summaries, figures, and SOP-like notes for internal review, strengthening CLIA-style validation readiness, traceability, and cross-functional collaboration between computational analysis and wet-lab assay development.

Bioinformatics Pipeline Developer | Boston University | *Boston, MA* | 2025-Present

- Designed RNA-seq workflows using FastQC, STAR, featureCounts, DESeq2, edgeR, limma, and Bioconductor for QC, alignment, quantification, differential expression, pathway interpretation, and host-response gene-signature review.
- Implemented motif discovery and promoter-analysis pipelines with MEME Suite, biomaRt, UCSC/Ensembl-style annotation resources, and Linux/HPC practices to support regulatory interpretation of high-throughput sequencing results.
- Built reproducible analysis notebooks and workflow documentation for collaborator-facing reruns, using Conda, Docker, Git, Snakemake/Nextflow concepts, and SLURM-aware execution patterns.
- Integrated expression matrices, annotation tables, and statistical outputs into clean deliverables for biological interpretation, publication-style figures, research summaries, and grant-supporting preliminary analyses.

AI Model Evaluation Analyst | Sepal AI | *Remote* | 2025-Present

- Evaluated 400+ model outputs across scientific reasoning, structured data transformation, conditional logic, and analytical workflows, building QA discipline directly applicable to computational pipeline validation and diagnostic bioinformatics support.
- Designed benchmark prompts and failure-mode reports for transformer-based systems, improving reproducibility, consistency, interpretability, and error detection across technical outputs.

AI/ML Research Analyst Intern | MBDA Federal Procurement Center | *Washington, DC* | Jun. 2023-Aug. 2023

- Built Python/pandas machine-learning analyses for large structured datasets, including feature engineering, model comparison, statistical visualization, and stakeholder-ready reporting.
- Authored a 10-page technical report on generative AI and transfer learning applications, connecting machine-learning methods to deployment, research strategy, and decision-support use cases.

SELECTED BIOINFORMATICS PROJECTS

Microbial Detection & Host-Response Multi-Omics Workflow | Python, R, Bioconductor, Snakemake, Docker

- Designed a Columbia CII-aligned workflow concept for integrating RNA-seq, high-throughput serology/protein-array outputs, microbiome-style feature tables, proteomics/metabolomics schemas, and clinical/sample metadata into analysis-ready matrices for infection, immunity, autism, and chronic fatigue syndrome/myalgic encephalomyelitis research contexts.
- Mapped outputs to microbial discovery and host-response questions: QC reports, differential-expression tables, pathway enrichment, biomarker ranking, assay-performance summaries, and reproducible reruns for shared HPC environments.

Clinical Genomics Dashboard for RNA-seq Analysis | Python, FastAPI, React, PostgreSQL, Docker

- Built REST APIs and a database-backed dashboard for querying differential expression results, gene annotations, volcano plots, and pathway enrichment outputs from FastQC/STAR/featureCounts/DESeq2 workflows.

Biomedical Literature RAG Assistant for Gene Interpretation | Python, LangChain, OpenAI API, FAISS

- Built a retrieval-augmented system for gene-function and disease-association interpretation, combining vector search over biomedical literature with structured summaries for researcher review.

AWARDS & RESEARCH WRITING

Graduate Tuition Merit Scholarship, Boston University (2025-Present); Dean's List, University of Washington (2020-2025). Authored publication-style internal technical report: Machine Learning and Deep Learning Advancements and Opportunities for Minority Business Enterprises, U.S. Department of Commerce MBDA (2023).